

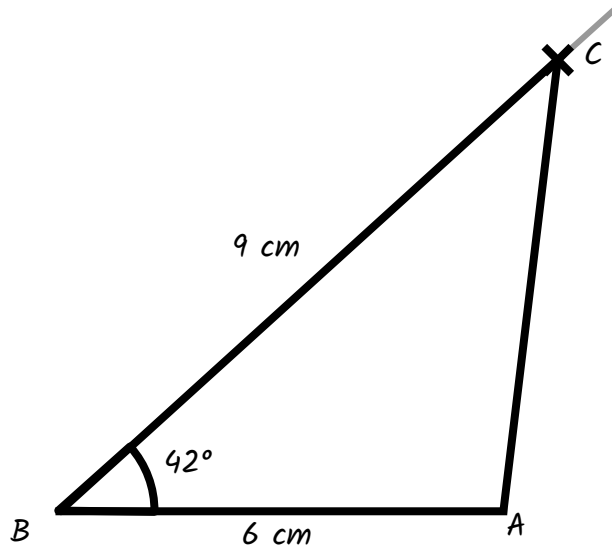


Congruent triangles (SAS)

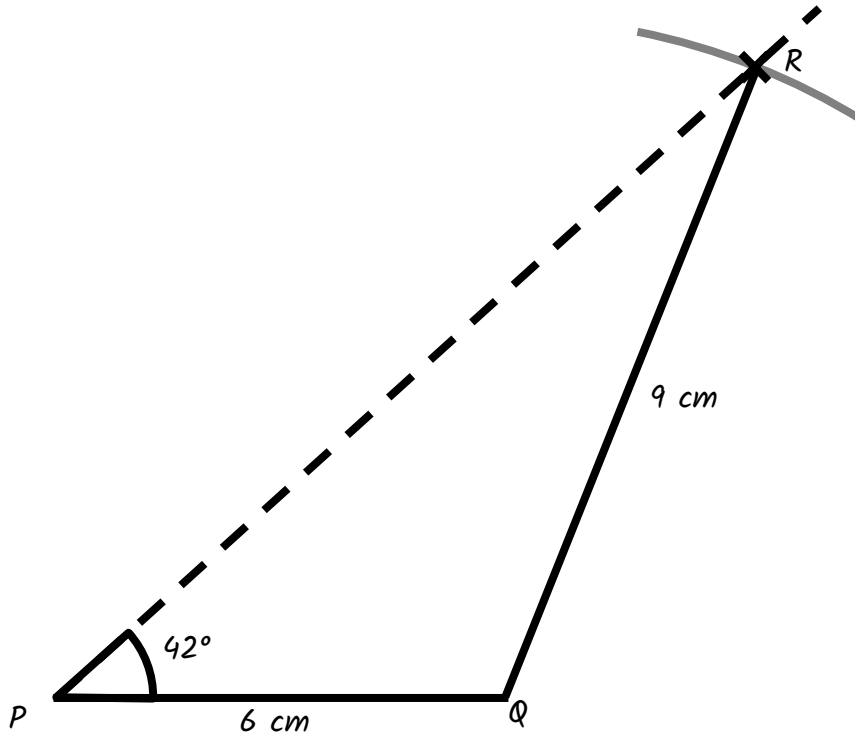
Task A: Justifying congruence by SAS

1) Construct the following triangles:

a) Triangle ABC, where $AB = 6\text{ cm}$, $BC = 9\text{ cm}$ and $\angle ABC = 42^\circ$



b) Triangle PQR, where $PQ = 6\text{ cm}$, $QR = 9\text{ cm}$ and $\angle QPR = 42^\circ$



2) Measure the edges and angles in the two triangles constructed in question 1. Are they congruent triangles?

Triangle ABC:

$$AB = 6 \text{ cm}$$

$$BC = 9 \text{ cm}$$

$$AC = 6.1 \text{ cm}$$

$$\angle ABC = 42^\circ$$

$$\angle BAC = 96.6^\circ$$

$$\angle ACB = 41.4^\circ$$

Triangle PQR:

$$PQ = 6 \text{ cm}$$

$$QR = 9 \text{ cm}$$

$$PR = 12.5 \text{ cm}$$

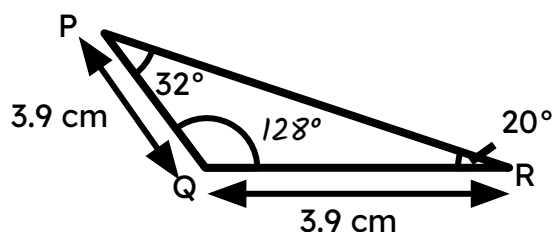
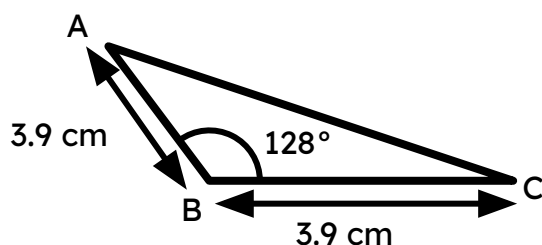
$$\angle QPR = 42^\circ$$

$$\angle PQR = 111.5^\circ$$

$$\angle PRQ = 26.5^\circ$$

No they are not congruent.

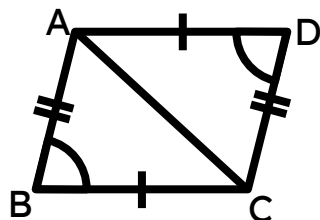
3) Prove triangle ABC is congruent to triangle PQR by SAS.



The interior angles of a triangle sum to 180° , therefore $\angle PQR = 180 - (32 + 20) = 128^\circ$.

$AB = PQ$, $\angle ABC = \angle PQR$ and $BC = QR$, so triangle ABC is congruent to triangle PQR by SAS (side-angle-side).

4) Given a parallelogram ABCD, prove that triangle ABC is congruent to triangle ACD by SAS.



$AD = BC$ as opposite edges on a parallelogram are equal in length.

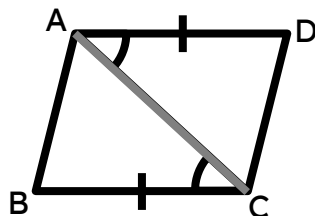
$\angle ABC = \angle ADC$ as opposite angles in a parallelogram are equal in size.

$AB = DC$ as opposite edges on a parallelogram are equal in length.

Hence, triangle ABC is congruent to triangle ACD by SAS.



4) Alternative answer:



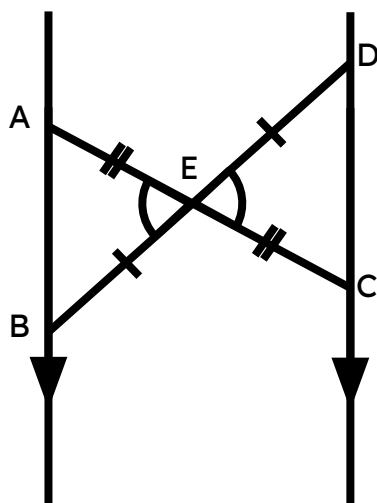
$AD = BC$ as opposite edges on a parallelogram are equal in length.

$\angle DAC = \angle ACB$ as they are equal interior alternate angles.

AC is a shared side.

Hence, triangle ABC is congruent to triangle ACD by SAS.

5) Point E is the midpoint of AC and BD . Prove that triangle AEB is congruent to triangle ECD by SAS.



$DE = EB$, as E is the midpoint of BD .

$\angle AEB = \angle DEC$ as vertically opposite angles.

$AE = EC$, as E is the midpoint of AC .

Hence, triangle AEB is congruent to triangle ECD by SAS.

Task B: Two sides and an angle

1) Which of the following triangles are congruent to the triangle XYZ.

